

Exploratory Data Analysis (EDA): OpenDengue Dataset (National Extract V1.3)

Project: OutbreakSignal DE — Nhóm Microwave (AIO 2026, Module 4: Data Sources & Ingestion)
Target: Dengue Surveillance in Southeast Asia (Ground-Truth Disease Records)
Dataset Reference: OpenDengue V1.3 ([National_extract_V1_3.csv](#))

Executive Summary

OpenDengue provides global, curated longitudinal epidemiological data compiled from WHO, Project Tycho, peer-reviewed literature, and national Ministries of Health. In this EDA, we analyze the **National Extract V1.3** to evaluate its suitability for the Bronze and Silver ingestion layers of OutbreakSignal DE.

High-Level Statistics

Metric	Global Dataset	Southeast Asia (SEA) Subset
Total Rows	29,873	3,579
Total Columns	16	16
Countries Covered	129 nations	11 nations (all ASEAN + Timor-Leste)
Temporal Span	1924 – 2025 (101 years)	1955 – 2025 (70 years)
Cumulative Recorded Cases	76,817,163	16,988,963 (~22.1% of global cases)
Spatial Resolution (S_res)	100% Admin0 (National)	100% Admin0 (National)
Temporal Resolutions (T_res)	Week (77.8%), Year (11.7%), Month (10.5%)	Week (59.4%), Month (27.7%), Year (12.9%)

1. Schema & Data Structure

The dataset contains 16 columns categorized into 4 semantic groups:

OpenDengueNationalRecord

```
+String adm_0_name
+String full_name
+String ISO_A0
+Int FAO_GAUL_code
+String RNE_iso_code
+Date calendar_start_date
+Date calendar_end_date
+Int Year
+Float dengue_total
+String case_definition_standardised
+String UUID
```

```
+Float adm_1_name(100% null)
+Float adm_2_name(100% null)
+Float IBGE_code(100% null)
+String T_res(Week, Month, Year)
+String S_res(Admin0)
```

Column Profiling

• Geographical Identifiers:

- **adm_0_name**: Canonical country name (e.g., **VIET NAM**, **THAILAND**, **PHILIPPINES**).
- **full_name**: Fully qualified administrative name.
- **ISO_A0**: ISO 3166-1 alpha-3 code (e.g., **VNM**, **THA**, **PHL**).
- **FAO_GAUL_code** & **RNE_iso_code**: Interoperability codes for UN FAO and regional mapping.
- **adm_1_name**, **adm_2_name**, **IBGE_code**: **100% Null (29,873 / 29,873 missing)**. Expected because this is the national-level aggregate extract (**S_res == Admin0**).

• Temporal Attributes:

- **calendar_start_date** & **calendar_end_date**: ISO format (**YYYY-MM-DD**).
- **Year**: Integer reporting year (1924 to 2025).
- **T_res**: Resolution of the reporting interval (**Week**, **Month**, **Year**).

• Epidemiological Metrics:

- **dengue_total**: Float count of cases recorded in the given interval.
- **case_definition_standardised**: Categorical classification of the diagnostic criteria.
- **S_res**: Spatial resolution (**Admin0** across entire file).
- **UUID**: Cryptographic/curation identifier referencing source paper, WHO bulletin, or Tycho stream.

2. Critical Data Engineering Traps & Quality Findings

During our deep audit, we discovered four subtle traps that directly impact the PySpark pipeline design:

[!WARNING]

Trap 1: The "VIET NAM" Whitespace Trap

In OpenDengue, Vietnam is officially recorded as "VIET NAM" (with a space), following UN/FAO conventions, **NOT** "VIETNAM".

A standard lookup `adm_0_name == "VIETNAM"` silently filters out all 317 Vietnam rows, underreporting the region by 4.56 million cases!

- **Fix for Silver Layer:** Standardize country strings using `trim()`, uppercase, and regex/alias mapping (`replace("VIET NAM", "VIETNAM")` or mapping to ISO `VNM`).

[!IMPORTANT]

Trap 2: Mixed Temporal Resolutions (`T_res`) in a Single Column

The column `dengue_total` contains **three heterogeneous time grains**:

- Some rows are 1-week counts (`T_res == 'Week'`)
- Some rows are 1-month counts (`T_res == 'Month'`)
- Some rows are 1-year annual totals (`T_res == 'Year'`)

Key Finding on Overlap: Within any given country and calendar year, OpenDengue uses **non-overlapping resolutions** (i.e. a country in year Y has either 52 weekly records, 12 monthly records, OR 1 annual record; never both).

Danger: Direct time-series modeling without resampling will compare a single week (~1,000 cases) to an entire annual total (~150,000 cases). Downstream Silver transformations must explicitly group by (`adm_0_name, Year`) for annual trends or resample to monthly grains.

[!NOTE]

Trap 3: Case Definition Standardization Quirks

The field `case_definition_standardised` contains:

- `Total` (23,592)
- `Suspected` (3,295)
- `Confirmed` (1,307)
- `Probable and confirmed` (1,074)
- `Suspected and confirmed` (399)
- `Probable` (103)
- `confirmed` (103) ← **Case mismatch:** lowercase "confirmed" vs capitalized "Confirmed".

In Southeast Asia, 97.4% of records are categorized as `Total`. In Silver, apply `lower(trim(case_definition_standardised))` to harmonize the labels.

[!TIP]

Trap 4: Zero Duplicate Records

We audited duplicates on composite key (adm_0_name, calendar_start_date, calendar_end_date).

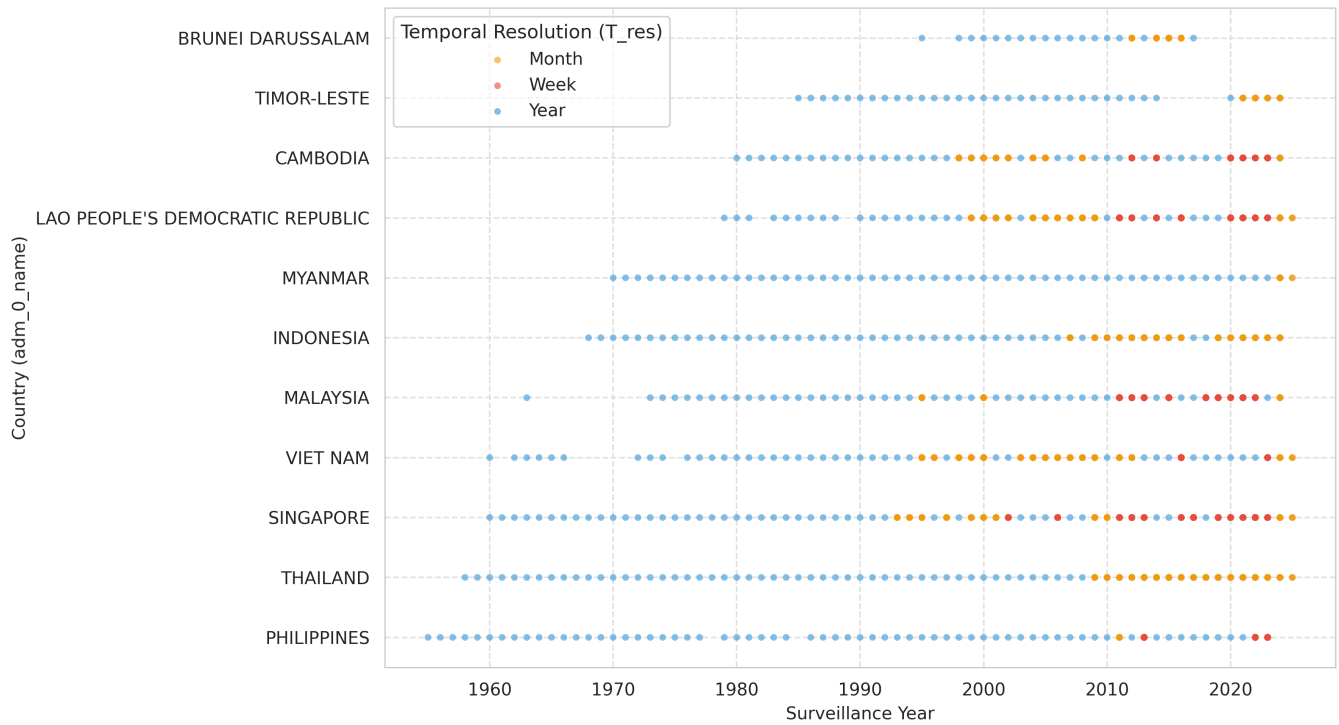
- **Global Duplicates:** 0
- **SEA Duplicates:** 0 This confirms high curation quality: OpenDengue does not have conflicting reports for the same date interval in the National extract.

3. Southeast Asia (SEA) Regional Coverage

OpenDengue provides continuous coverage for all 11 nations in Southeast Asia:

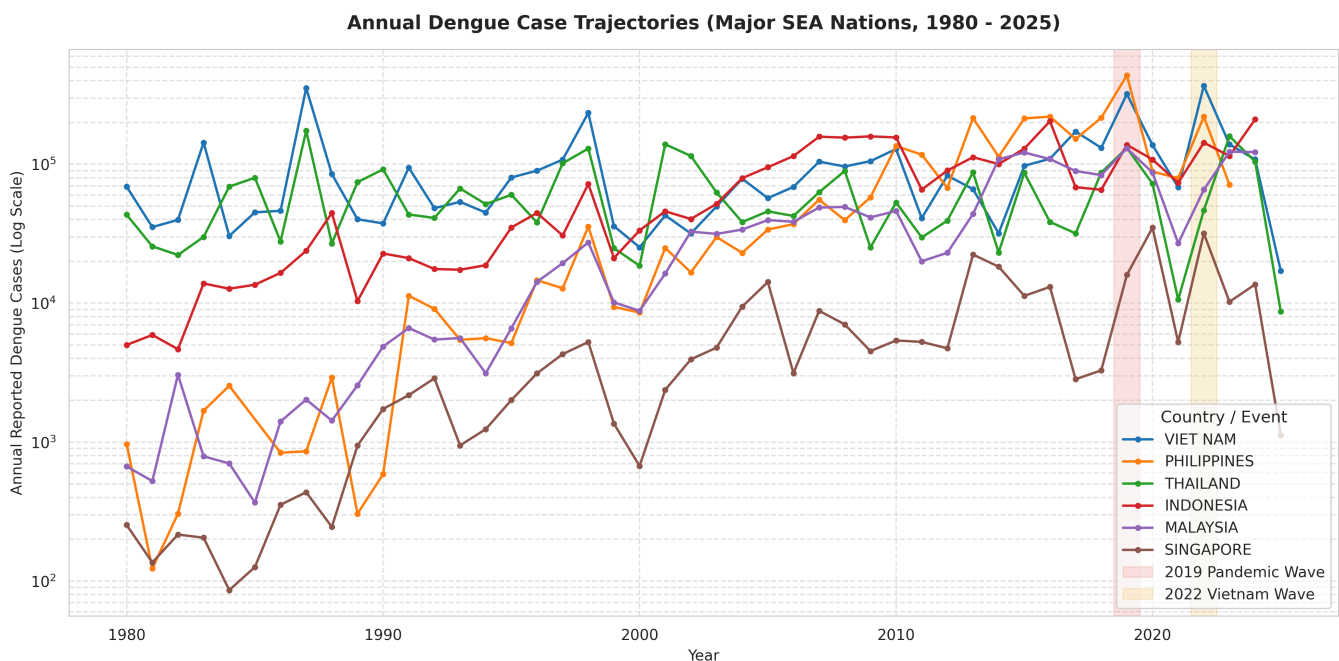
Country (adm_0_name)	ISO	Total Records	Surveillance Range	Week Rows	Month Rows	Year Rows	Cumulative Cases
VIET NAM	VNM	317	1960 – 2025	93	183	41	4,565,891
INDONESIA	IDN	220	1968 – 2024	0	178	42	3,212,124
THAILAND	THA	246	1958 – 2025	0	195	51	3,068,213
PHILIPPINES	PHL	224	1955 – 2023	149	12	63	2,822,549
MALAYSIA	MYS	545	1963 – 2024	468	36	41	1,667,648
CAMBODIA	KHM	449	1980 – 2024	311	108	30	511,978
MYANMAR	MMR	68	1970 – 2025	0	14	54	488,835
LAO PDR	LAO	575	1979 – 2025	415	135	25	354,180
SINGAPORE	SGP	793	1960 – 2025	627	123	43	292,770
TIMOR-LESTE	TLS	77	1985 – 2024	0	46	31	16,265
BRUNEI	BRN	65	1995 – 2017	0	48	17	4,005
TOTAL	—	3,579	1955 – 2025	2,063	978	538	16,988,963

OpenDengue V1.3: Surveillance Timeline & Temporal Resolution (Southeast Asia)



4. Epidemiological Trends & Historic Outbreaks

By aggregating each country's records by (`adm_0_name`, `Year`), we reconstruct the continuous annual caseload across the region from 1980 to 2025.



Key Outbreak Waves in Southeast Asia:

1. The 2019 Pan-Regional Dengue Emergency:

- A super-epidemic swept across Southeast Asia:
 - **Philippines:** 437,563 cases (highest recorded in national history, national dengue epidemic declared).

- **Vietnam:** 320,702 cases.
- **Malaysia:** 130,101 cases.
- **Thailand:** 131,157 cases.

2. The 2022 Vietnam Wave:

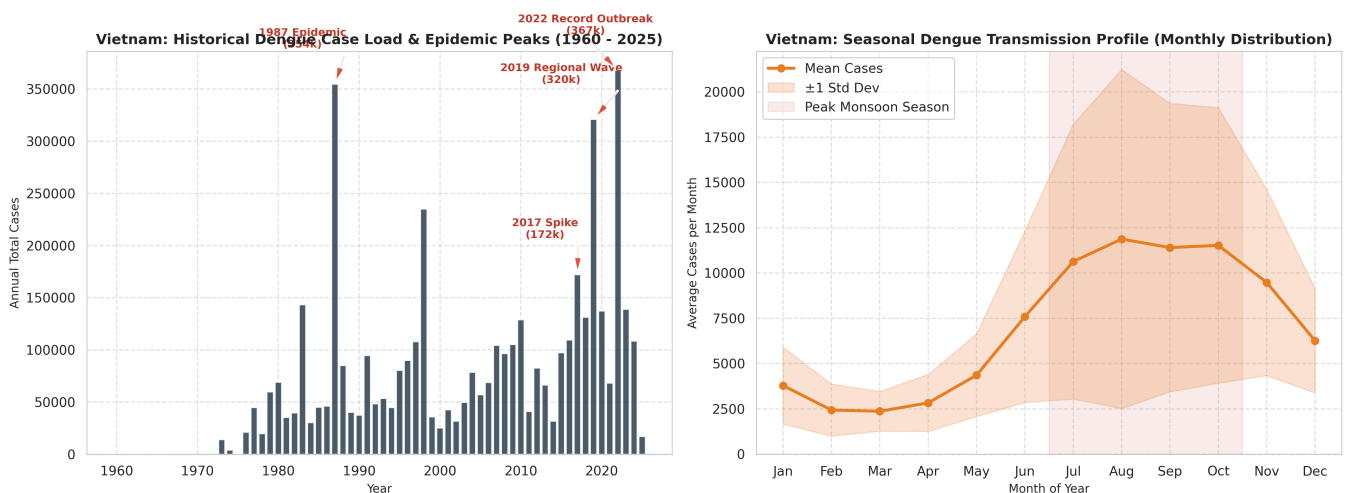
- Vietnam suffered an all-time record outbreak of **367,729 cases**, surpassing the 2019 peak.

3. The 1987 Historical Epidemic:

- Severe dengue hemorrhagic fever epidemic across mainland SEA: Vietnam recorded **354,517 cases** and Thailand recorded **174,285 cases**.

5. Vietnam In-Depth Outbreak & Seasonality Analysis

Vietnam represents the single largest cumulative case burden in Southeast Asia within OpenDengue (over 4.56 million recorded cases).



Seasonal Dynamics in Vietnam

From the monthly surveillance subset ($T_{res} == \text{'Month'}$), dengue incidence demonstrates strong seasonal periodicity:

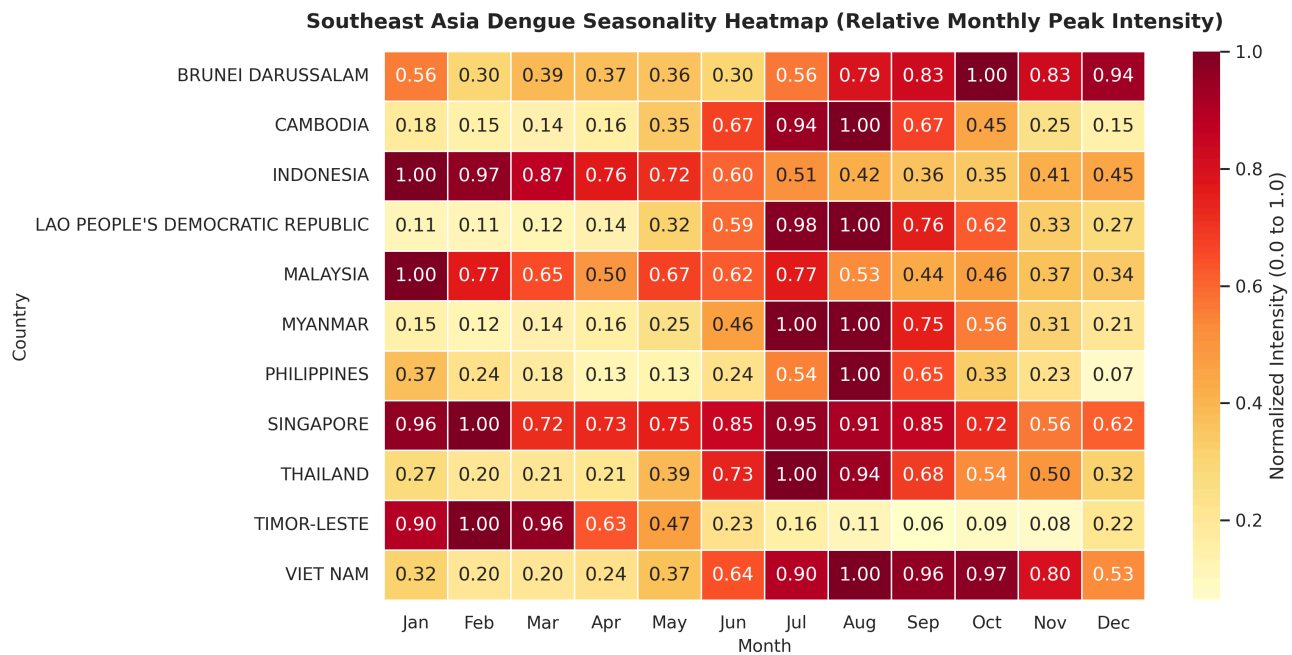
- **Baseline / Low Season (January – April):** Average monthly incidence remains stable between **2,300 and 3,700 cases/month**.
- **Onset / Acceleration (May – June):** Cases rapidly climb from **4,358 (May)** to **7,590 (June)** as the Southwest monsoon sets in.
- **Peak Epidemic Season (July – October):** Monthly caseload stays above **10,500 – 11,872 cases/month**, peaking in August/October.
- **Tapering (November – December):** Gradual decrease to **9,476 (Nov)** and **6,261 (Dec)**.

[!TIP] Outbreak Early-Warning Rule for OutbreakSignal DE:

A reported monthly case count in Vietnam exceeding **5,000 cases before June** or a weekly count exceeding **1,500 cases in Q1/Q2** indicates an abnormal early outbreak wave, which should trigger high-priority alerts when correlated with spikes in dengue-related Google News articles.

6. Comparative Regional Seasonality

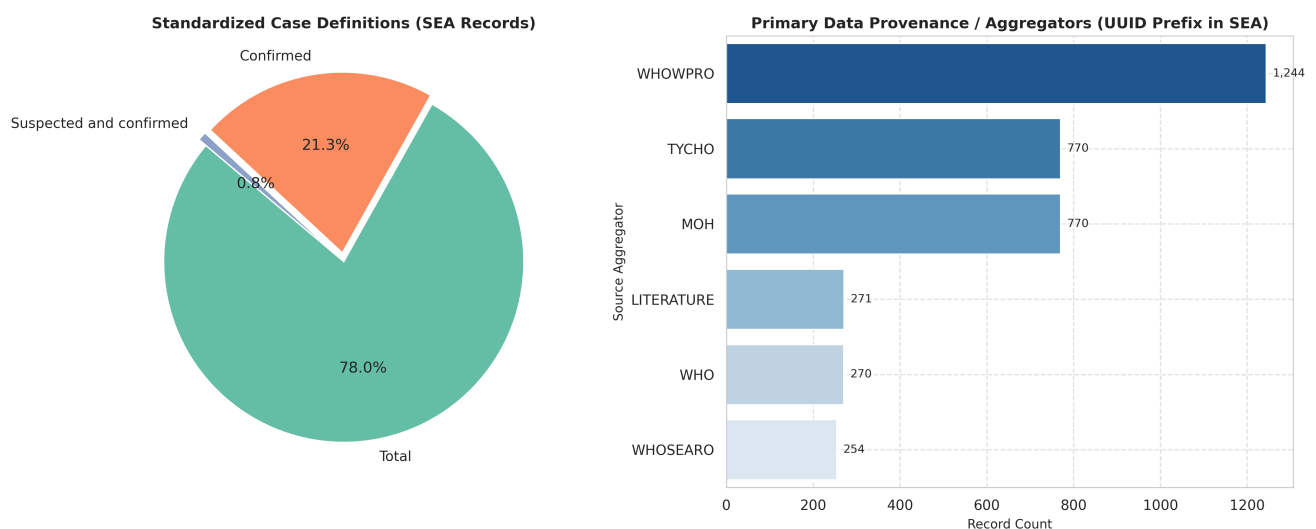
The transmission season depends heavily on geography and latitude across Southeast Asia:



- **Northern Tropical Belt (Vietnam, Cambodia, Laos, Thailand):**
 - Peak intensity occurs between **June and October** (coninciding with the rainy/monsoon season).
- **Southern Hemisphere Tropical Belt (Timor-Leste):**
 - Peak transmission is **inverted**: highest cases occur between **December and March** (Southern tropical rainy season), while July to October are minimal.

7. Data Provenance & Case Definitions

Understanding data provenance ensures that downstream data lake layers trace data lineage back to trustworthy primary sources.



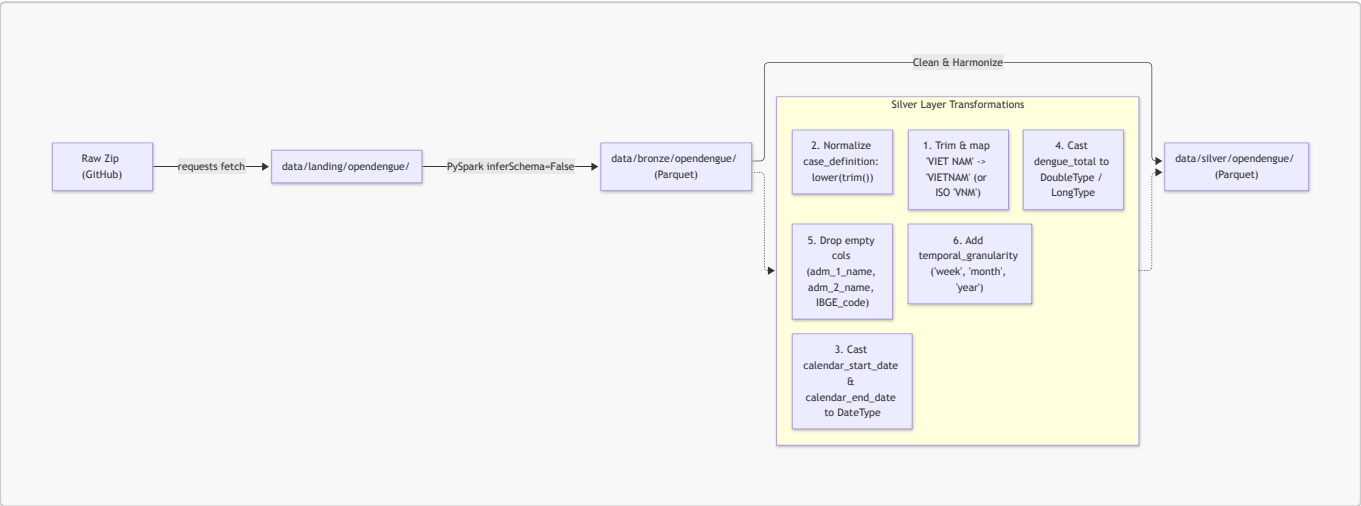
Primary Curators (UUID Prefixes):

1. **TYCHO (Project Tycho):** Provides historical digitized surveillance (1960s – 2010s).

2. **WHOWPRO (WHO Western Pacific Regional Office):** Supplies weekly and monthly official surveillance bulletins for Vietnam, Cambodia, Laos, Philippines, Singapore, Malaysia.
3. **WHO (Global Health Observatory):** Annual historical statistics.
4. **LITERATURE:** Curated peer-reviewed epidemiological research papers filling historical gaps.

8. Recommendations for Pipeline Implementation (Bronze & Silver)

Based on these findings, here is the architectural specification for the OutbreakSignal DE pipeline:



1. **Bronze Layer:**
- Adhere strictly to the bronze invariant: Keep `inferSchema=False`, do not drop null columns, do not normalize country names. Store with partition `ingest_date=YYYY-MM-DD`.
2. **Silver Layer:**
- Create a clean standardized table with:
 - Standardized country: `country_name` (VIETNAM), `country_iso` (VNM).
 - Cast dates to `DateType` and case counts to `IntegerType` / `DoubleType`.
 - Cleaned `case_definition` in lowercase.
 - Drop the 100% null columns (`adm_1_name`, `adm_2_name`, `IBGE_code`).
 - Expose both the raw interval records and a monthly-resampled view for easy correlation with news signals.

9. Deliverables in Codebase

File	Description
<code>[src/eda_opendengue.py]</code> (file:///Users/iamqazolp/Documents/Projects/microwave/outbreak-signal-de/src/eda_opendengue.py)	Standalone automated Python EDA pipeline script.
<code>[notebooks/eda_opendengue.ipynb]</code> (file:///Users/iamqazolp/Documents/Projects/microwave/outbreak-signal-de/notebooks/eda_opendengue.ipynb)	Interactive 17-cell Jupyter Notebook for exploratory execution.

File	Description
[spikes/test_opendengue.py] (file:///Users/iamqazolp/Documents/Projects/microwave/outbreak-signal-de/spikes/test_opendengue.py#L29-L44)	Updated spike test including 'VIETNAM' country filter.
[docs/images/*.png] (file:///Users/iamqazolp/Documents/Projects/microwave/outbreak-signal-de/docs/images)	5 publication-ready 300-DPI visualization figures.